

Homework 1

CS 09410 and CS 09510

Topics: Chapter 1

Due February 8, 11.59 pm

Spring 2026

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Read the following instructions before submitting your work:

- Submitting work copied (in part or in whole) from other persons, websites, or generative AI tools such as ChatGPT is strictly forbidden.
- Submit your answers on Canvas as a single PDF file.
- Include a heading with your name and assignment number/date.
- Handwriting is allowed if writing is readable by others.

Remarks: 1 byte is 8 bits. bps stands for bits per second; Bps, with uppercase B, stands for bytes per second. Letters k, M, and G denote powers of 10: k is 10^3 , M is 10^6 , and G is 10^9 . That is, 1 kBps = 8 kbps = 8,000 bps.

Problem 1: Describe the most popular wireless *access* technologies today. Compare and contrast them.

Problem 2: Suppose there is exactly one packet switch between a sending host and a receiving host. The transmission rates between the sending host and the switch and between the switch and the receiving host are R_1 and R_2 , respectively. If the switch uses store-and-forward packet switching, what is the total end-to-end *transmission* delay for a packet of length L bits?

Problem 3: Suppose users share a 2 Mbps link; each user transmits continuously at 1 Mbps when transmitting, but each user transmits only 20% of the time.

- a) When circuit switching is used, how many users can be supported?
- b) Assume packet switching is used. Will there be any significant queuing delay if two or fewer users transmit simultaneously? What if three users transmit simultaneously?
- c) Suppose now there are three users. Find the probability that at any given time, one user is transmitting.
- d) Now find the probability that all three users are transmitting simultaneously.

Problem 4: Consider sending a packet from a source host to a destination host over a fixed route. List the delay components in the end-to-end delay. Which of these delays tend to be constant and which tend to be variable?

Problem 5: How long does it take to transmit a packet of length 1,000 bytes over a link of distance 2,500 km, propagation speed 2.5×10^8 m/sec, and transmission rate 2 Mbps?

Problem 6: Suppose Host A wants to send a large file to Host B. The path from A to B has three links of rates $R_1 = 500$ kbps, $R_2 = 2$ Mbps, and $R_3 = 1$ Mbps.

- a) Assuming no other traffic in the network, what is the throughput for the file transfer?
- b) What if R_2 is reduced to 100 kbps?

Problem 7: What are the five layers in the Internet protocol stack? Briefly describe the main responsibilities of each of these layers.